Deepfake Audio Detection: Real vs. Synthetically Cloned Voices

Muhammad Ahmad
25280065

Israr Ali
25280102

Imran Saeed
25280083

***Abstract*—Deepfake audio is becoming increasingly realistic, making it difficult to distinguish between genuine and fake speech using traditional detection methods. This project proposes a robust deepfake audio detection framework focused on improving generalization against unseen spoofing attacks. The system uses the ASVspoof 2019 Logical Access dataset, where the training set contains known attacks and the evaluation set contains mostly unknown attacks, making it suitable for real-world testing.**

The proposed architecture begins with audio preprocessing, including mono conversion, 16 kHz resampling, waveform normalization, variable-length handling, padding, and masking. To improve robustness, RawBoost and codec augmentation are applied during training to simulate real-world distortions such as channel noise, signal corruption, and compressed audio. A strong teacher model is then trained using Wav2Vec2-XLSR as the front-end feature extractor and AASIST as the back-end classifier. The teacher model learns deep speech representations and detects spectral-temporal spoofing artifacts.

To further improve generalization, knowledge distillation is used. The trained teacher model generates soft predictions, which are used to train a lighter student model consisting of a frozen Wav2Vec2-XLSR front-end and AASIST-L back-end. The student model learns from both the original labels and the teacher's soft predictions using a combination of hard label loss and knowledge distillation loss. The final system classifies speech as either bona fide or spoofed.

The model is evaluated using EER, min-tDCF, accuracy, precision, recall, F1-score, and confusion matrix. This project aims to build a practical and reliable deepfake audio detection system that performs well not only on benchmark data but also on unseen and distorted audio conditions.

I. Problem Statement

Deepfake audio has become increasingly realistic due to rapid advances in speech synthesis and voice conversion technologies. This creates a serious challenge for voice-based security systems, online communication, and digital media verification. Although many existing detection models achieve high accuracy on benchmark datasets, they often fail to generalize when tested on unseen spoofing attacks, noisy recordings, or compressed real-world audio. This project addresses this problem by developing a robust deepfake audio detection framework that combines strong speech feature extraction, audio augmentation, and knowledge distillation to improve detection performance against unknown and real-world spoofed speech.

II. Literature Review

Speech deepfake detection has become an important research area because modern voice conversion and speech synthesis systems can generate highly realistic fake speech. The main problem is not only detecting fake audio on benchmark datasets, but also building models that can generalize to unseen attacks, noisy audio, compressed speech, and real-world conditions. This section reviews ten important papers related to the ASVspoof 2019 Logical Access task and explains how each paper supports the proposed methodology.

A. Survey on Speech Deepfake Detection

Li et al. [1] presented a broad survey of speech deepfake detection methods. This paper is important because it does not propose one single model, but summarizes the strongest trends across more than 200 studies. The survey shows that recent high-performing systems often combine self-supervised learning front-ends, strong anti-spoofing back-ends, advanced loss functions, and augmentation techniques.

The paper identifies a strong general recipe for speech deepfake detection: wav2vec 2.0/XLSR as the front-end, AASIST as the back-end, OC-Softmax as the loss function, and RawBoost or codec augmentation for robustness. It also highlights that models can achieve very low EER on ASVspoof 2019 LA, but their performance often drops during cross-dataset testing. This means that generalization is still the main challenge. For our project, this paper supports the use of Wav2Vec2-XLSR, AASIST, RawBoost, codec augmentation, OC-Softmax, and knowledge distillation.

B. ASVspoof 2019 Database

Wang et al. [2] introduced the ASVspoof 2019 database, which is selected as the main dataset for this research. The Logical Access track contains both bona fide and spoofed speech samples. The training and development sets include known attacks A01–A06, while the evaluation set contains mostly unknown attacks A07–A19. This makes the dataset suitable for testing generalization because the model must detect spoofing attacks that were not seen during training.

The paper also provides two official baseline systems: CQCC-GMM and LFCC-GMM. The LFCC-GMM baseline performs better, achieving 8.09% EER and 0.2116 min-tDCF on the LA evaluation set, while CQCC-GMM achieves 9.57% EER and 0.2366 min-tDCF. Therefore, LFCC-GMM is selected as the baseline model in this project. It provides a traditional machine learning reference point for comparing the proposed deep learning framework.

C. ASVspoof 2019 Challenge Results

Todisco et al. [3] reported the official ASVspoof 2019 challenge results. This paper is useful because it shows the performance gap between traditional baselines and modern neural-network-based systems. The winning Logical Access system, T05, achieved 0.22% EER and 0.0069 min-tDCF, which was significantly better than the official baselines.

The paper also reports that the top-performing systems mainly used neural networks and ensemble fusion. However, some attacks, especially A10, A13, and A17, remained difficult. This shows that even strong systems can struggle with specific spoofing methods. For our project, this paper supports the use of deep learning models and motivates the need for a generalization-focused architecture rather than a simple single-model approach.

D. End-to-End Anti-Spoofing with RawNet2

Tak et al. [4] proposed RawNet2, an end-to-end anti-spoofing model that directly processes raw waveform input. The model uses fixed sinc filters, residual blocks, filter-wise attention, a GRU layer, and a final fully connected classifier. RawNet2 achieved 4.66% EER and 0.1294 min-tDCF on ASVspoof 2019 LA.

This paper is important because it shows that raw waveform models can learn useful spoofing patterns without relying completely on handcrafted features such as LFCC or CQCC. The fusion of LFCC-GMM and RawNet2 achieved 1.12% EER, showing that combining traditional and deep learning systems can improve performance. In our project, RawNet2 is considered a useful deep learning baseline, but AASIST is selected as the main back-end because it provides stronger spoofing detection performance.

E. AASIST: Spectro-Temporal Graph Attention

Jung et al. [5] introduced AASIST, a graph-attention-based model for audio anti-spoofing. AASIST uses a RawNet2-style encoder followed by spectral and temporal graph attention modules. These modules help the model capture spoofing artifacts from both frequency and time domains. AASIST achieved 0.83% EER and 0.0275 min-tDCF on ASVspoof 2019 LA, making it one of the strongest single systems.

The paper also introduced AASIST-L, a lightweight version with only 85K parameters, which achieved 0.99% EER. This is important for our methodology because the full AASIST model can be used as the teacher model, while AASIST-L can be used as the student model in knowledge distillation. Therefore, this paper directly supports the back-end design of our proposed architecture.

F. Comparative Study on Neural Spoofing Countermeasures

Wang and Yamagishi [6] compared several neural countermeasure systems for synthetic speech detection. They proposed the LCNN-LSTM-sum model with P2SGrad loss. Their best system achieved 1.92% EER and 0.057 min-tDCF on ASVspoof 2019 LA. The study also showed that results can vary across different random seeds, with EER changing from 1.92% to 3.10% across multiple runs.

This paper is important for two reasons. First, it supports the use of P2SGrad as a strong loss function for spoofing detection. Second, it shows that single-run results are not always reliable. Therefore, our methodology includes multi-seed evaluation so that the final results are more stable and trustworthy. This paper also supports the idea of comparing different loss functions such as P2SGrad and OC-Softmax.

G. wav2vec 2.0 Self-Supervised Speech Representations

Baevski et al. [7] introduced wav2vec 2.0, a self-supervised learning model for speech representation learning. Although wav2vec 2.0 was originally developed for automatic speech recognition, it is highly useful for speech deepfake detection because it learns strong speech representations from large amounts of unlabeled audio.

The model contains a CNN feature encoder and a Transformer-based context network. It learns from raw audio using contrastive learning and masking. For our project, wav2vec 2.0/XLSR is used as the front-end feature extractor. Instead of relying only on handcrafted features, Wav2Vec2-XLSR provides deep speech representations that can help improve generalization against unseen spoofing attacks.

H. SSL Front-Ends for Spoofing Countermeasures

Wang and Yamagishi [8] investigated self-supervised learning front-ends for spoofing countermeasures. Their best system used W2V-XLSR with an LLGF back-end and achieved 0.11% EER on ASVspoof 2019 LA. The model also performed well on cross-condition datasets, showing that multilingual SSL front-ends can improve generalization.

The paper found that fine-tuning W2V-XLSR gives much better results than using it as a frozen feature extractor. However, fine-tuning requires more computational resources and careful training. In our project, Wav2Vec2-XLSR is selected as the main front-end because it provides strong self-supervised speech features. The teacher model can be fine-tuned for stronger performance, while the student model can use a frozen Wav2Vec2-XLSR front-end for efficiency.

I. RawBoost Data Augmentation

Tak et al. [9] proposed RawBoost, a raw waveform augmentation method designed for anti-spoofing. RawBoost simulates real-world distortions such as channel effects, non-linear distortion, and signal-dependent noise. The best configuration combines linear/non-linear convolutive distortion with impulsive signal-dependent noise. This setup reduced EER from 9.50% to 5.31% on ASVspoof 2021 LA.

This paper is highly relevant because our project focuses on generalization. Real-world audio may contain compression, channel noise, device distortion, and other signal changes. RawBoost helps the model become more robust to such conditions. Therefore, RawBoost 1+2 is selected as the main augmentation technique in our proposed methodology. Codec augmentation is also added to simulate compressed audio from platforms such as WhatsApp, Zoom, and social media.

J. Does Audio Deepfake Detection Generalize?

Müller et al. [10] studied whether audio deepfake detection models generalize to real-world data. They evaluated several architectures trained on ASVspoof 2019 LA and tested them on both ASVspoof evaluation data and an in-the-wild dataset. The results showed a major performance drop. For example, RawGAT-ST achieved 1.23% EER on ASVspoof but degraded to 37.15% EER on in-the-wild data.

This paper strongly supports the main motivation of our project. It shows that low EER on ASVspoof does not always mean the model will work well in real-world conditions. The paper also shows that variable-length input can perform better than fixed 4-second truncation, and that experiments should be repeated using multiple random seeds. Therefore, our methodology includes variable-length handling, padding and masking, augmentation, knowledge distillation, and multi-seed evaluation to improve generalization.

K. Summary of Reviewed Literature

The reviewed literature shows a clear progression from traditional handcrafted-feature systems to modern deep learning and self-supervised learning methods. The key findings from the literature and their role in the proposed methodology are summarized below.

- **Traditional Baseline Methods:** CQCC-GMM and LFCC-GMM are traditional spoofing detection methods based on handcrafted acoustic features. Among them, LFCC-GMM performs better on ASVspoof 2019 LA and is therefore selected as the baseline model for this project.
- **Raw Waveform Learning:** RawNet2 demonstrates that spoofing patterns can be learned directly from raw audio waveforms. This supports the use of end-to-end deep learning models instead of relying only on handcrafted features.
- **Graph-Attention Back-End:** AASIST captures spectral and temporal spoofing artifacts using graph attention. Based on its strong performance, AASIST is selected as the teacher back-end, while AASIST-L is used as the lightweight student back-end.
- **Self-Supervised Front-End:** Wav2Vec2-XLSR provides strong self-supervised speech representations learned from large-scale speech data. It is used as the front-end feature extractor in the proposed architecture.
- **Data Augmentation:** RawBoost and codec augmentation improve robustness against noise, channel distortion, and compressed audio. These augmentations are applied during training to improve generalization.
- **Loss Functions:** P2SGrad and OC-Softmax improve class separation between bona fide and spoofed speech. These losses are considered for training the teacher model.
- **Knowledge Distillation:** Knowledge distillation transfers useful knowledge from a strong teacher model to a smaller student model. In this project, the teacher model provides soft predictions to train the student model.
- **Generalization Evaluation:** Literature shows that low EER on ASVspoof does not always guarantee good performance on unseen or in-the-wild attacks. Therefore, the final system is evaluated using EER, min-tDCF, accuracy, precision, recall, F1-score, and confusion matrix.
- **Proposed Direction:** Based on these findings, this project proposes a knowledge-distilled Wav2Vec2-XLSR-AASIST framework using RawBoost, codec augmentation, and a lightweight student model for improved generalization.

III. Methodology

This research proposes a simple and robust deepfake audio detection methodology. The main goal is to detect whether an audio sample is real, called bona fide, or fake, called spoof. The proposed method uses the ASVspoof 2019 Logical Access dataset, audio preprocessing, training augmentation, a teacher model, a student model, and knowledge distillation.

A. Dataset

The ASVspoof 2019 Logical Access dataset is used in this research. It contains both real and spoofed speech samples. The training and development sets contain known spoofing attacks, while the evaluation set contains mostly unknown attacks. This makes the dataset useful for testing how well the model can detect unseen deepfake audio.

B. Preprocessing

Before training, all audio files are prepared in the same format. First, the audio is converted into mono format. Then, it is resampled to 16 kHz. After that, waveform normalization is applied to reduce loudness differences between audio samples. Since all audio files do not have the same duration, variable-length handling is used. Padding and masking are also applied so that audio files can be processed in batches without losing important speech information.

C. Training Augmentation

Data augmentation is applied during training to make the model more robust. RawBoost augmentation is used to add realistic distortions to the audio. In this project, RawBoost + is used. RawBoost adds linear and non-linear convolutive distortion, while RawBoost adds impulsive signal-dependent noise. Codec augmentation is also used to simulate compressed audio, such as audio sent through calls, social media, or messaging applications. These augmentations help the model perform better on noisy and real-world audio.

D. Teacher Model

The teacher model is the stronger model in the proposed system. It uses Wav2Vec2-XLSR as the front-end and AASIST as the back-end. Wav2Vec2-XLSR extracts deep speech features from the audio, while AASIST classifies the audio as bona fide or spoof by learning spectral and temporal spoofing patterns. The teacher model is trained using P2SGrad or OC-Softmax loss to improve the separation between real and fake speech.

E. Teacher Soft Predictions

After the teacher model is trained, it produces soft predictions for the training samples. These soft predictions show how confident the teacher model is about each class. For example, instead of only saying that an audio is spoof, the teacher may give scores such as 0.92 for spoof and 0.08 for bona fide. These soft predictions contain useful information and are used to train the student model.

F. Student Model

The student model is a lighter model. It uses a frozen Wav2Vec2-XLSR front-end and AASIST-L as the back-end. The Wav2Vec2-XLSR front-end is kept frozen, which means its weights are not updated during student training. It is only used for feature extraction. AASIST-L is trained as the classifier because it is a lightweight version of AASIST.

G. Knowledge Distillation

Knowledge distillation is used to transfer knowledge from the teacher model to the student model. During training, the student model learns from two things: the original hard labels and the teacher's soft predictions. The hard label loss helps the student learn the correct class, while the knowledge distillation loss helps it learn from the teacher's confidence scores. This helps the student model become more accurate and generalize better to unseen spoofing attacks.

H. Final Prediction

After training, the student model is used as the final prediction model. It takes an input audio sample and predicts whether it is bona fide or spoof. Bona fide means the audio is real, while spoof means the audio is fake or deepfake.

I. Evaluation

The final model is evaluated using both standard classification metrics and anti-spoofing metrics. The main metrics are EER and min-tDCF because they are commonly used in ASVspoof research. Accuracy, precision, recall, F1-score, and confusion matrix are also used to understand the model performance in more detail.

IV. Exploratory Data Analysis

Before starting model training, an exploratory data analysis (EDA) was performed on the ASVspoof 2019 Logical Access (LA) dataset. The purpose of this step was to understand the dataset structure, verify the protocol files, inspect the class distribution, check speaker separation, and confirm that the audio files were correctly linked with their metadata. This analysis helped ensure that the dataset was clean and suitable for deepfake audio detection.

A. Dataset Loading and Protocol Understanding

The ASVspoof 2019 LA dataset was loaded from the raw dataset directory. It contains three main splits: training, development, and evaluation. The protocol files were used to read the metadata of each audio sample. Since the protocol files do not contain column names, the columns were first loaded as generic fields and then renamed for clarity. The final columns used in the analysis were *speaker_id*, *audio_id*, *system_id*, *attack_type*, and *label*. The label column identifies whether the audio sample is *bonafide* or *spoof*.

TABLE I
Total number of samples in each dataset split.

Dataset Split	Total Samples
Train	25,380
Development	24,844
Evaluation	71,237

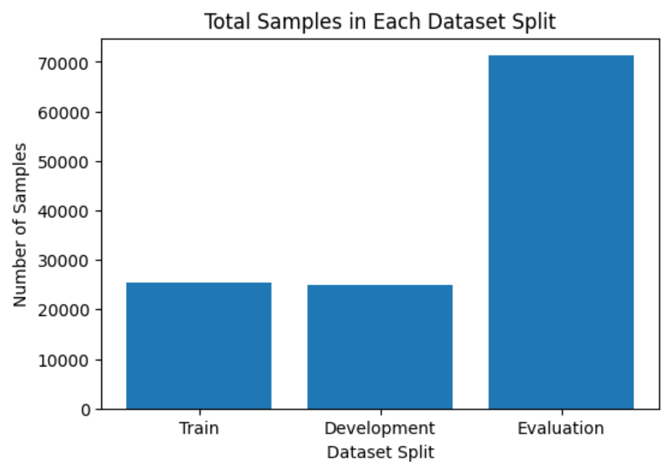

Fig. 1. Total samples in each dataset split.

B. Label Distribution

The dataset contains two classes: *bonafide*, which represents real human speech, and *spoof*, which represents fake or synthetically generated speech. The label distribution shows that the dataset is highly imbalanced, with spoof samples forming almost 90% of each split. This class imbalance is important because it can influence model training and evaluation. Therefore, metrics such as EER, min-tDCF, precision, recall, and F1-score are more meaningful than accuracy alone.

TABLE II
Bonafide and spoof sample distribution.

Label	Train	Development	Evaluation
Spoof	22,800	22,296	63,882
Bonafide	2,580	2,548	7,355

TABLE III
LABEL PERCENTAGE IN EACH DATASET SPLIT.

Label	Train	Development	Evaluation
Spoof	89.83%	89.74%	89.68%
Bonafide	10.17%	10.26%	10.32%

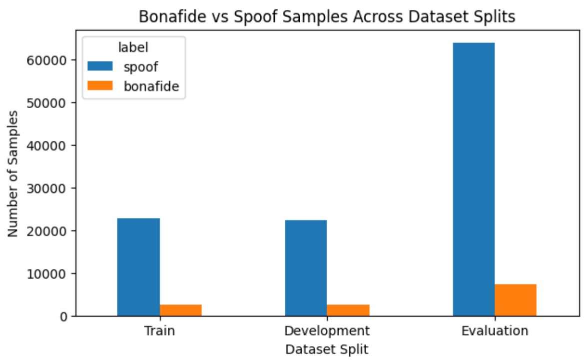

Fig. 2. Bonafide and spoof samples across dataset splits.

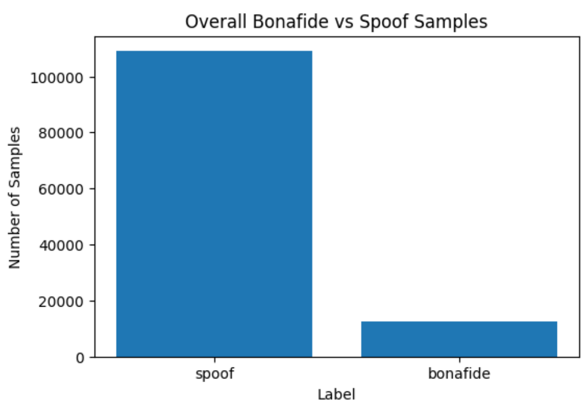

Fig. 3. Overall bonafide and spoof sample count.

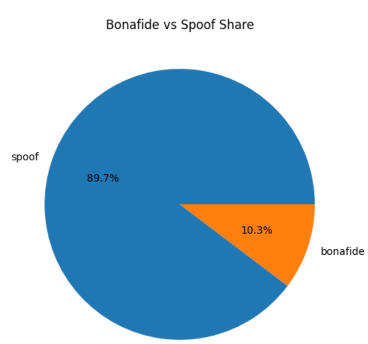

Fig. 4. Overall percentage share of bonafide and spoof samples.

C. Speaker Distribution and Speaker Overlap

The number of unique speakers was also analyzed for each split. The training set contains 20 unique speakers, the development set contains 20 unique speakers, and the evaluation set contains 67 unique speakers. Speaker overlap was checked between the splits to ensure that the same speaker does not appear in multiple splits. The result showed zero overlap between train and development, train and evaluation, and development and evaluation sets. This confirms that the dataset is properly separated and supports fair evaluation on unseen speakers.

TABLE IV
UNIQUE SPEAKERS IN EACH DATASET SPLIT.

Dataset Split	Unique Speakers
Train	20
Development	20
Evaluation	67

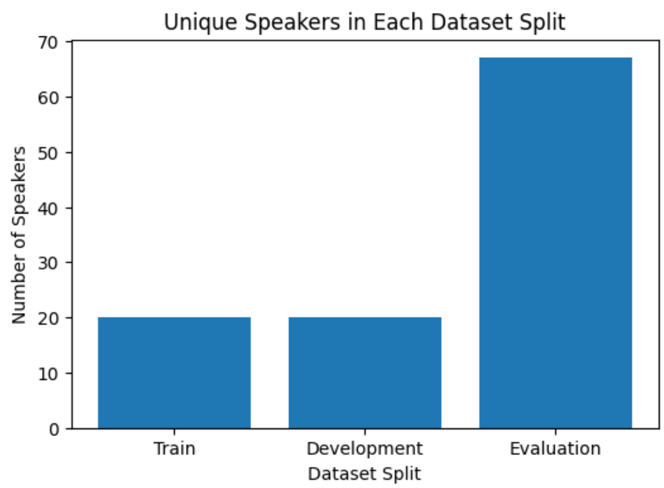

Fig. 5. Unique speakers in each dataset split.

D. Attack Type and Spoofing System Analysis

The *attack_type* column indicates which spoofing attack was used to generate a spoofed sample. For bonafide samples, this value is represented by a dash because no spoofing attack is applied. The distribution of attack types was examined in the training set to understand the types of attacks available during model learning. Similarly, the *system_id* column was analyzed to observe which spoofing systems were used to generate the fake audio samples. These analyses are useful because the model must learn general spoofing patterns rather than memorizing one specific attack or system.

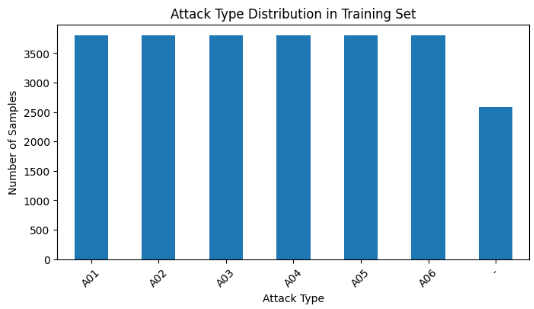

Fig. 6. Attack type distribution in the training set.

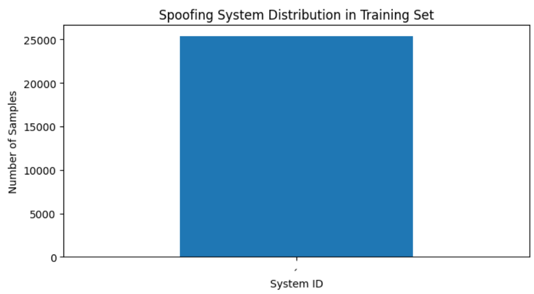

Fig. 7. Spoofing system distribution in the training set.

E. Missing Values and Audio File Verification

Missing value analysis was performed on the training, development, and evaluation protocol files. No missing values were found in any of the main columns. After this, full audio file paths were generated by combining each audio ID with its corresponding folder path and the `.flac` extension. The number of audio files was compared with the number of protocol rows. The training set had an exact match between protocol rows and audio files. The development and evaluation folders contained more audio files than the protocol rows, but all protocol records were successfully linked for analysis.

TABLE V
PROTOCOL ROWS AND AVAILABLE AUDIO FILES.

Dataset Split	Protocol Rows	Audio Files
Train	25,380	25,380
Development	24,844	24,986
Evaluation	71,237	71,933

F. EDA Summary

The exploratory data analysis confirmed that the ASVspoof 2019 LA dataset is suitable for the proposed deepfake audio detection system. The dataset contains a large number of spoof and bonafide audio samples, has clearly defined train, development, and evaluation splits, and does not contain missing metadata values. The analysis also showed that the classes are imbalanced, with spoof samples being much more frequent than bonafide samples. Speaker overlap between the splits was zero, which supports reliable testing on unseen speakers. Overall, this EDA provided a clear understanding of the dataset before preprocessing, augmentation, model training, and evaluation.

V. Future Work

In the next submission, the proposed deepfake audio detection system will be further improved and extended. First, the complete preprocessing pipeline will be finalized, including mono conversion, 16 kHz resampling, waveform normalization, padding, and masking. After this, training augmentation techniques such as RawBoost and codec augmentation will be applied to make the model more robust against noisy and real-world audio conditions.

The next step will be to train the teacher model using Wav2Vec2-XLSR as the front-end and AASIST as the back-end. The teacher model will be trained using suitable anti-spoofing loss functions such as P2SGrad or OC-Softmax. After training, the teacher model will generate soft predictions for the training samples. These soft predictions will then be used to train a lighter student model through knowledge distillation.

In the student model, the Wav2Vec2-XLSR front-end will be frozen and AASIST-L will be used as a lightweight back-end classifier. The student model will learn from both the original hard labels and the teacher model's soft predictions. This step is expected to improve the model's generalization ability while reducing model complexity.

Finally, the trained student model will be evaluated using standard anti-spoofing and classification metrics. The main evaluation metrics will include Equal Error Rate (EER), minimum Tandem Detection Cost Function (min-tDCF), accuracy, precision, recall, F1-score, and confusion matrix. The results will be compared with the teacher model to analyze the performance difference between the large model and the lightweight distilled model. In future work, the system can also be tested on real-world audio samples collected from online calls, social media, and messaging platforms to further evaluate its practical performance.

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